

A robust, general-purpose K_{DP} algorithm for LROSE. A constrained self-consistency approach for S-, C-, and X-band radars.

Extended Abstract

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Abstract

Reliable estimation of specific differential phase (K_{DP}) from measured differential phase (Φ_{DP}) requires careful treatment of measurement noise, phase folding, differential backscatter phase, and, particularly at C- and X-band, attenuation. We present a robust general-purpose K_{DP} algorithm intended as a documented reference implementation for the Lidar Radar Open Software Environment (LROSE).

The method first identifies regions containing usable Φ_{DP} , performs phase unfolding, and applies spatial filtering in range. A key feature is the subsequent application of self-consistency to individually selected segments of the filtered Φ_{DP} profile. The segments are chosen to contain localized increasing–decreasing phase excursions that may include differential backscatter phase. Within each segment, the Z_H – Z_{DR} self-consistency relationship determines the relative distribution of propagation phase, while the filtered Φ_{DP} values at the segment endpoints constrain the total phase change. This endpoint-constrained formulation preserves the observed large-scale phase structure and limits the influence of the self-consistency reconstruction to the local segment. Differential backscatter phase (δ) is subsequently estimated from the difference between the filtered measured phase and the reconstructed propagation phase. Attenuation corrections to Z_H and Z_{DR} are included iteratively where appropriate.

The algorithm has been tested using S-, C-, and X-band weather radar observations. Initial results show consistent behavior across the three radar bands and demonstrate that the segmented self-consistency procedure can preserve the measured phase structure while redistributing propagation phase toward regions supported by the polarimetric measurements. The method will be released as open-source software within LROSE.

Acknowledgements

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1. Introduction

1.1. Goal

Specific differential phase (K_{DP}) is a quantity derived from the measured differential phase (Φ_{DP}). The measurements are quite noisy and require significant filtering for reliable results. A significant number of methods have been developed for estimating K_{DP} , leaving users confused about which method to use. Adding to this confusion is the fact that K_{DP} is dependent on radar wavelength.

Aldana et al. (2025) provide an excellent survey of the available methods, and for which purpose they should be used.

The goal of this research is to develop a method, based on existing methods and techniques, that is robust and applies to all of the common use cases and radar wavelengths. The method will be provided in the open source Lidar Radar Open Software Environment (LROSE) repository on GitHub, making it available to all users. A later goal is to port the method to Python in the PyArt radar package.

1.2. Notes on the data sets and cases

We want the proposed method to work reliably with all weather radar wavelengths. From a practical point of view, for precipitation detection and estimation, we need to support S-band (10 cm), C-band (5 cm), and X-band (3 cm). So we tested the algorithm on radars from all bands. For this paper, we selected a suitable case from each of the bands. These are detailed in Table 1.

Table 1: Data set details and sources

Band	Radar name	λ (cm)	Date	Time	Organization
S	S-Pol	10.68	2015/06/26	00:08:31	NSF-NCAR, PECAN field campaign
C	COW2	5.5	2025/06/18	01:12:11	UAH-FARM, ICECHIP field campaign
X	DOW7	3.2	2025/06/18	00:52:02	UAH-FARM, ICECHIP field campaign

1.3. Theoretical overview

As a dual-polarization radar wave propagates through a volume of hydrometeor particles, the wave experiences a progressive phase delay. The delay is larger for particles with larger dimensions. Since some particles are not spherical, the delay in one polarization may be different from that in the other. This results in a non-zero measured differential phase Φ_{DP} .

Measured differential phase Φ_{DP} can be expressed as (Hubbert and Bringi (1995); Wang and Chandrasekar (2009); Aldana et al. (2025)):

$$\Phi_{DP}(r) = \Psi_{DP}(r) + \delta(r) + \epsilon(r), \quad (1)$$

where Ψ_{DP} is the accumulated differential propagation phase, δ is the backscatter differential phase, and ϵ represents measurement noise and other errors. Ideally we would like to determine the values of Ψ_{DP} and δ separately - that is the goal of most algorithms and techniques devoted to estimating K_{DP} from Φ_{DP} .

The specific differential phase K_{DP} is related to the propagation phase by

$$K_{DP}(r) = \frac{1}{2} \frac{d\Psi_{DP}(r)}{dr}. \quad (2)$$

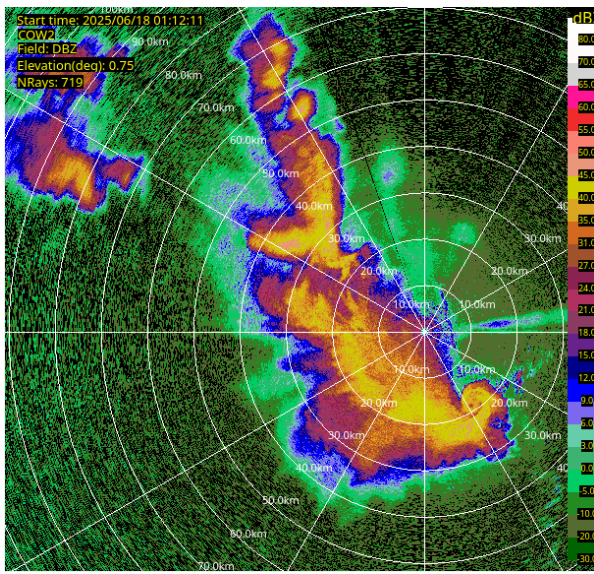
In other words, K_{DP} is linearly related to the rate of phase change with range along the beam.

2. Example observations

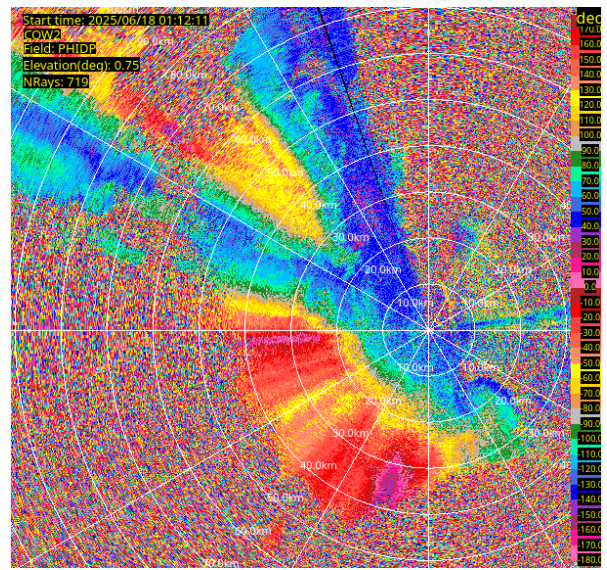
Φ_{DP} is generally measured as the difference between the H-channel phase and the V-channel phase, with units of degrees. For a simultaneous H/V transmit radar (i.e. the most common) Φ_{DP} wraps at values of -180 and +180. For an alternating-mode radar, the wrap occurs at -90 and +90 degrees.

A dual-polarization radar has a 'system Φ_{DP} ' which remains largely constant. This system H-V difference is the result of the specific hardware in the radar - i.e. transmit/receive paths and the waveguide and cable lengths. We can estimate the system Φ_{DP} by observing echo close to the radar, say in light stratiform rain in which the drops are essentially spherical. As the beam propagates and traverses regions containing hydrometeors, Φ_{DP} generally increases, though it is observed to decrease sometimes.

In regions of good signal Φ_{DP} is smoothly varying, though with a gate-to-gate variability related to measurement noise. In regions without returned signal Φ_{DP} is effectively random, fluctuating between the folding limits.



(a) Reflectivity from storm case
COW2 C-band radar, 2025/06/18.



(b) Φ_{DP} for the C-band storm case in Figure 1a.

Figure 1: Example of C-band reflectivity and Φ_{DP} .

Figure 1a shows the reflectivity from a Doppler On Wheels C-Band radar for a strong storm case with a hook echo. Figure 1b shows the measured Φ_{DP} for this case. As you can see, the Φ_{DP} signature is coherent in regions with reasonable signal-to-noise ratio (SNR), but essentially random where there is no significant return. As we work through the KDP algorithm, we will distinguish between the regions with good Φ_{DP} and the regions with noise.

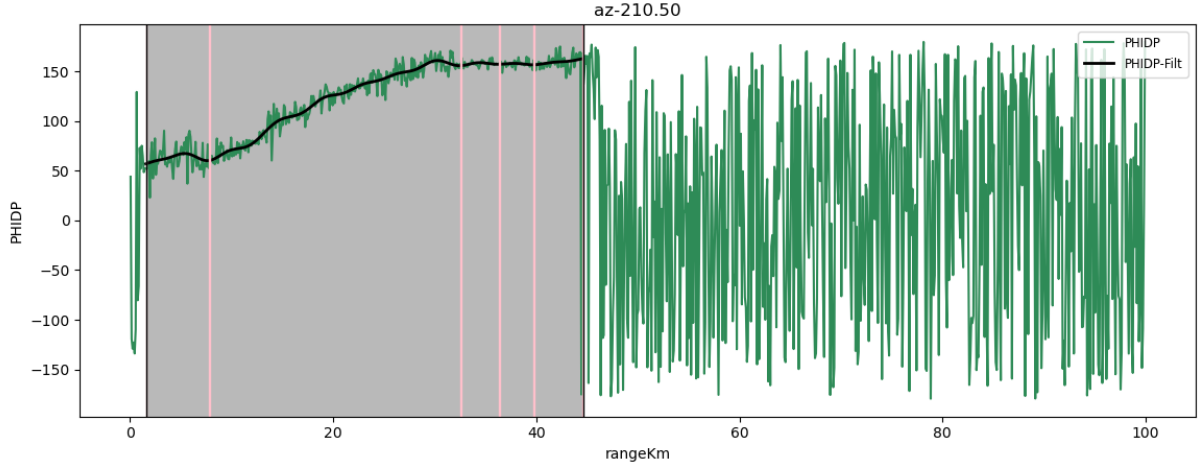


Figure 2: Ray plot of Φ_{DP} , azimuth 210.5 deg, for case in Figure 1b. Green is unfiltered Φ_{DP} . The black line shows an example of range-filtered Φ_{DP} .

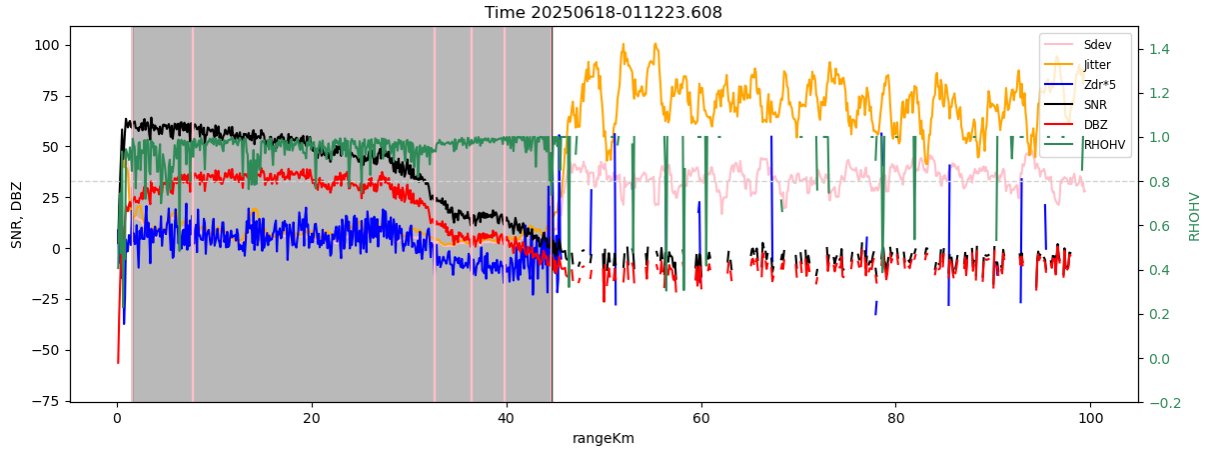


Figure 3: Ray plot of radar moments, azimuth 210.5 deg, for case in Figure 1b. Red - reflectivity. Black - SNR. Green - RHOHV. Blue - ZDR * 5. Pink - standard deviation of Φ_{DP} with range. Orange - jitter of Φ_{DP} with range.

Figure 2 shows a plot of Φ_{DP} with range, at an azimuth of 210.5 degrees, for the data in Figure 1b. The green trace is the measured Φ_{DP} . You can see that the Φ_{DP} field has low variance in the region with good signal (highlighted in gray), while Φ_{DP} is essentially random between -180 and +180 in the region further from the radar without echo. Figure 3 shows the moments for the same ray.

3. Identifying regions with valid Φ_{DP}

Refer again to Figures 2 and 3. The gray regions indicate the ranges with usable Φ_{DP} signatures.

To identify these usable regions we make use of 4 features, and apply thresholds to determine whether the features support valid KDP:

- The spatial standard deviation of Φ_{DP} in range.
- The spatial 'jitter' of Φ_{DP} in range.
- The signal-to-noise ratio SNR.
- The H-V cross-correlation coefficient ρ_{HV} .

This is similar to the design used by the Clutter Mitigation Decision algorithm: Hubbert et al. (2009).

We compute the spatial standard deviation and jitter over a specified number of gates - the computational kernel.

3.1. The spatial standard deviation of Φ_{DP}

Since Φ_{DP} is an angle, and can wrap, we need to take this into account when computing the standard deviation. We represent Φ_{DP} as a unit vector, associated with a point on the unit circle. We compute the conventional circular mean/resultant-vector statistic over the kernel, and then compute the standard deviation centered on that circular mean.

3.2. The spatial jitter of Φ_{DP}

'Jitter' as we define it is closely related to the standard deviation, but it is distinct and it provides some additional skill. Jitter is the mean angle change from gate to gate, measured around the unit circle. This is computed over the same kernel as the standard deviation.

3.3. Typical parameter settings

We have found that a kernel of 9 gates is suitable for these computations.

Table 2 shows typical values that work well for this step:

Table 2: Thresholds for determining valid Φ_{DP}

Feature field	Units	Threshold
Standard deviation	deg	20
Jitter	deg	30
SNR	dB	0
ρ_{HV}		0.85

4. Unfolding Φ_{DP}

As mentioned previously, Φ_{DP} will fold at $-180/+180$ degrees for simultaneous transmit radars, and $-90/+90$ degrees for alternating mode radars. We need to take care of this folding behavior to ensure that the algorithm works properly.

Unfolding is relatively straightforward if we treat Φ_{DP} as a unit vector. Consider Figure 4. This shows 2 vectors, one on each side of the fold at $-180/+180$:

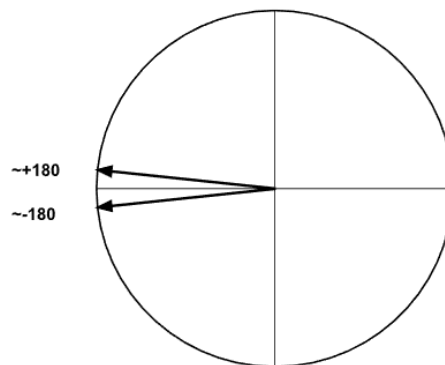


Figure 4: 2 Φ_{DP} vectors on either side of a fold.

We can detect a fold with the following logic:

- The X coordinates of both vectors are negative
- AND
- The Y coordinates of the vectors have opposite sign.

If this logic holds true, then a fold has occurred. If the Y coordinate of the second vector is positive, then the fold is in the sense of from -180 to +180. If the Y coordinate of the second vector is negative, then the fold is in the sense of from +180 to -180.

Before performing the unfolding step, we first compute the mean Φ_{DP} , using the same 9-gate kernel used to compute the standard deviation. And once again we perform the computations in vector space to avoid the folding problem we are seeking to mitigate.

At the end of this step, we have a ray of mean Φ_{DP} values that have been unfolded, and are therefore numerically contiguous.

For radars with a folding interval from -90 to +90 degrees, for this unfolding step we multiply Φ_{DP} by 2, perform the unfolding assuming a -180 to +180 degree interval, and then rescale by dividing by 2.

5. Filtering Φ_{DP} in range

Take another look at Figure 2. In the gray box we are showing the unfiltered Φ_{DP} values in green, overlain with a filtered version in black. It is the filtered Φ_{DP} that we discuss in this section. We will consider four options for filtering, and then discuss the advantages and disadvantages.

5.1. FIR filtering: Hubbert and Bringi (1995)

Hubbert and Bringi introduced a method for filtering Φ_{DP} using an FIR filter. The FIR approach has the advantage of being efficient and local, immune from influences some distance away from the range gate in question. A potential disadvantage is that the FIR filter must be designed beforehand, resulting in a choice of discrete filters from which we need to select. Nevertheless this is a well tested approach and has been used operationally for many years.

5.2. Quadratic polynomial filtering: Wang and Chandrasekar (2009)

The method of Wang and Chandrasekar (2009) relies on a local polynomial fit to Φ_{DP} , which both produces a smoothed version of Φ_{DP} and leads to a method for computing KDP from the polynomial fit. In this work, we test a version with a polynomial of order 2, i.e. a quadratic. We fit a local quadratic to the Φ_{DP} on either side of a selected range gate. From that quadratic we compute KDP as the slope of the fit at the center point. We then integrate KDP to produce filtered Φ_{DP} .

5.3. Higher-order polynomial filtering: Hubbert et al. (2021)

Hubbert et al. (2021) apply higher-order polynomial regression to the problem of ground clutter filtering. In this work, we use a similar approach to fit the observed Φ_{DP} in the regions of valid Φ_{DP} - i.e. we do not include the regions between the blocks of valid Φ_{DP} data.

5.4. FFT low-pass filtering

In this filter, the Φ_{DP} profile is smoothed in range using an FFT-based low-pass spatial filter, which suppresses small-scale range variability while retaining the larger-scale phase structure. This is implemented using a cut-off frequency in cycles/km. The FFT-based filter is applied to the entire ray. At gates without valid Φ_{DP} the value is interpolated from the adjacent valid regions.

5.5. Setting the filter parameters

To permit meaningful comparison among the filtering methods, each filter is parameterized in terms of a common physical quantity, termed the feature length, L_f . This represents the approximate minimum range scale of phase features that the filtering procedure is intended to preserve.

For the cases we have tested so far, a value of 3.5 km for L_f works well.

Table 3 shows how we choose the filter parameters.

Table 3: Parameters for the filter options

Filter type	Parameter setting
FIR	Use lookup table to match filter coefficients to L_f
Quadratic	Range of quadratic $X = L_f$
Regression	Polynomial order = (Spatial extent of data / L_f) * 2 + 1
FFT	Spatial cutoff frequency (cycles/km) = (1 / L_f)

5.6. Comparing the filters

Figure 5 shows an example of a ray plot in which all of the filters are plotted on top of the unfiltered Φ_{DP} . For the cases we have considered so far, we found that by setting the parameters carefully we were able to achieve similar results from each of the filters. This suggests that, for the cases examined here, the choice of filter is less important provided that the parameters are set to represent comparable spatial scales.

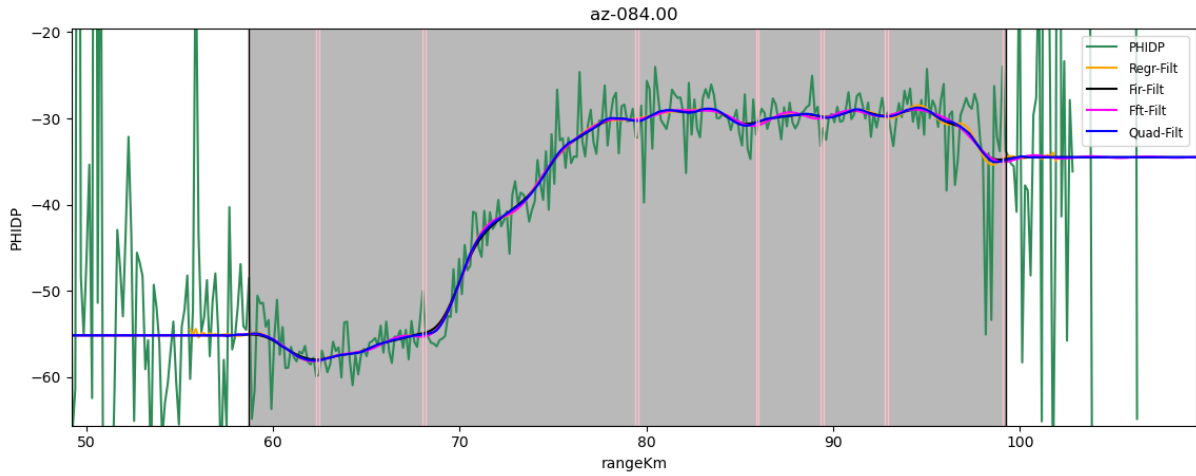


Figure 5: Ray plot showing the 4 filters. NCAR S-Pol S-band data example.

Figure 6 shows a zoomed view. You can see differences between the results of the filters. However, the differences are small for this example.

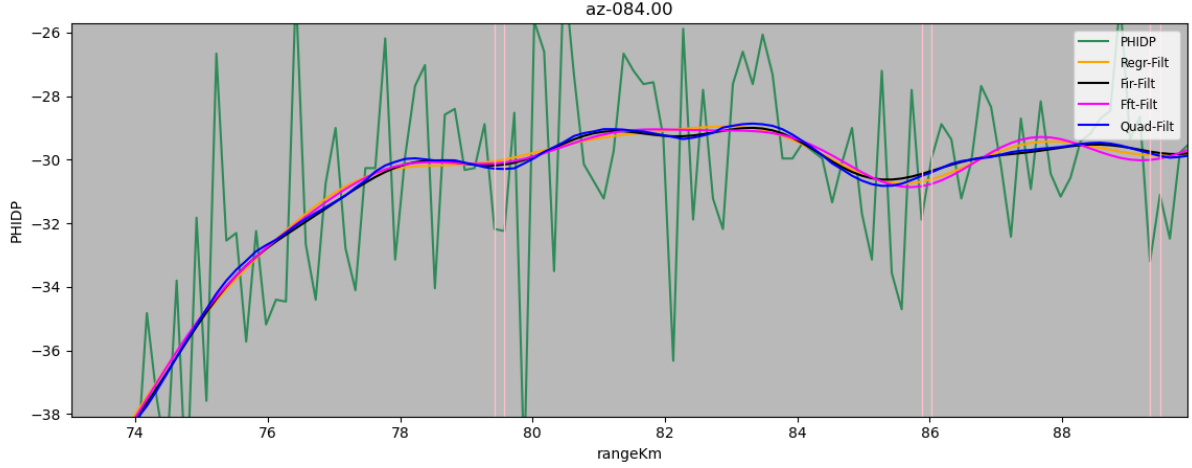


Figure 6: Zoomed ray plot comparing the 4 filters.

6. Identifying backscatter differential phase (δ)

Trömel et al. (2013) (Figure 2, not shown here) is an example of δ at X-band in a rain cell, causing elevated values of Φ_{DP} of up to 10 degrees above the background values to which Φ_{DP} returns on the far side of the cell. Such signatures are frequently seen at S- and C-band as well.

Figure 7 shows an example of probable backscatter differential phase at S-band, highlighted in yellow, for a case from the NCAR S-Pol radar. We would like to be able to reliably identify these signatures, and correct for them, so that the residual Ψ_{DP} can be used for such purposes as precipitation estimation, hydrometeor classification, etc.

Trömel et al. (2013) introduce a method for identifying δ that utilizes modified versions of the ZPHI method from Testud et al. (2000) and the self-consistency method of Bringi and Chandrasekar (2001). Our method builds on this framework, but uses a locally segmented, endpoint-constrained self-consistency reconstruction for individual segments in the d_P profile.

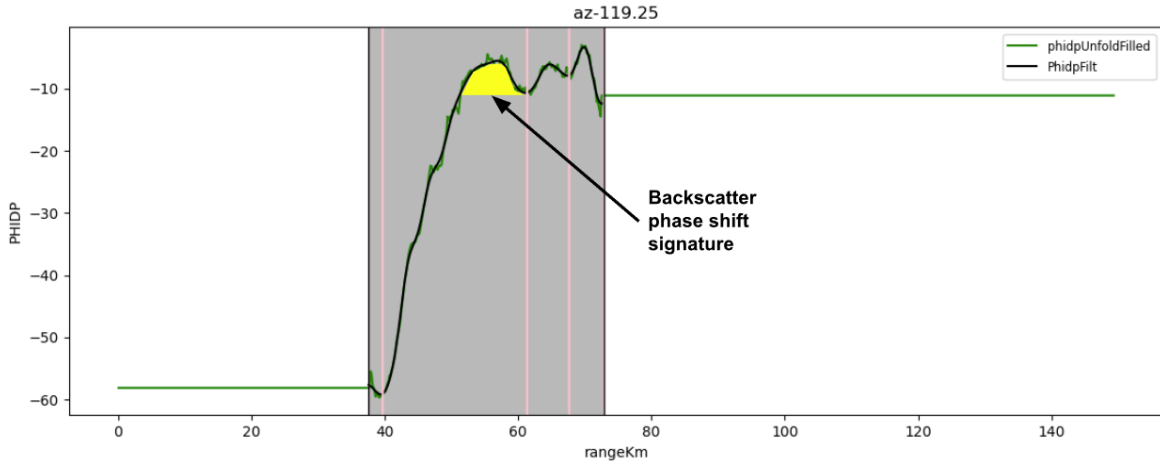


Figure 7: Backscatter differential phase signature (yellow).

In this work we follow a similar procedure to that described in Trömel et al. (2013).

6.1. The self-consistency method

The self-consistency method (Bringi and Chandrasekar (2001)) allows us to estimate K_{DP} from Z and Z_{DR} . We will refer to the derived quantity as K_{DP}^{sc} .

Scarchilli et al. (1996) give a self-consistency equation of the form:

$$K_{DP}^{sc} = a Z_H^b Z_{DR}^c, \quad (3)$$

where Z_H and Z_{DR} are expressed in linear units, a , b , and c are empirically derived coefficients, and K_{DP}^{sc} is the self-consistency estimate of specific differential phase in degrees km^{-1} . Here Z_h denotes horizontal reflectivity in dBZ, while Z_H denotes the corresponding linear reflectivity factor. Similarly, Z_{dr} denotes differential reflectivity in dB and Z_{DR} denotes the corresponding linear ratio:

$$Z_H = 10^{Z_h/10}, \quad Z_{DR} = 10^{Z_{dr}/10}. \quad (4)$$

From the literature, we have the following recommended parameters:

Table 4: Reference coefficients for the self-consistency relationship in Eq. (3).

Band	λ (cm)	a	b	c	Reference
S	10.0	0.0001051	0.96	-0.260	Scarchilli et al. (1996)
C	5.5	0.0001461	0.98	-0.200	Scarchilli et al. (1996)
X	3.2	0.000222	1.0	-4.58	Shi et al. (2018)

6.2. Correcting for attenuation

It is important, especially at C- and X-band, to account for attenuation effects.

In Table 7.1 of Bringi and Chandrasekar (2001), the attenuation relationships are commonly expressed as power laws involving specific differential phase K_{DP} :

$$A_H = a_H K_{DP}^{b_H}, \quad (5)$$

$$A_{DP} = A_H - A_V = a_{DP} K_{DP}^{b_{DP}}, \quad (6)$$

where A_H is specific attenuation at horizontal polarization in dB km^{-1} , A_V is the corresponding vertical-polarization attenuation, and A_{DP} is specific differential attenuation in dB km^{-1} .

Table 5 lists the reference parameters from Bringi and Chandrasekar (2001):

Table 5: Reference coefficients for attenuation correction.

Band	λ (cm)	a_H	b_H	a_{DP}	b_{DP}
S	10.0	0.017	0.84	0.003	1.05
C	5.5	0.073	0.99	0.013	1.23
X	3.2	0.233	1.02	0.033	1.15

The attenuation-corrected reflectivity and differential reflectivity, expressed in dB units, are

$$Z_h^{\text{corr}}(r) = Z_h^{\text{meas}}(r) + 2 \int_{r_0}^r A_H(s) ds, \quad (7)$$

and

$$Z_{dr}^{\text{corr}}(r) = Z_{dr}^{\text{meas}}(r) + 2 \int_{r_0}^r A_{DP}(s) ds, \quad (8)$$

where A_H is the one-way specific attenuation at horizontal polarization and $A_{DP} = A_H - A_V$ is the one-way specific differential attenuation, both in dB km^{-1} .

For use in the self-consistency relationship, the corrected quantities are converted to linear units:

$$Z_H^{\text{corr}}(r) = 10^{Z_H^{\text{corr}}(r)/10}, \quad (9)$$

and

$$Z_{DR}^{\text{corr}}(r) = 10^{Z_{DR}^{\text{corr}}(r)/10}. \quad (10)$$

The self-consistency equation may then be written as:

$$K_{DP}^{\text{sc}}(r) = a [Z_H^{\text{corr}}(r)]^b [Z_{DR}^{\text{corr}}(r)]^c. \quad (11)$$

6.3. Segmenting - selecting suitable range limits

A feature of the self-consistency method is that it does not affect the overall phase change in a profile, but rather redistributes the phase changes to where they are most appropriate. For that to work reliably we need to identify suitable short segments of the profile over which we apply the method. A useful technique here is to identify selected sections of the filtered Φ_{DP} profile that are first monotonically increasing, and then monotonically decreasing, before increasing again.

These boundaries are chosen because a localized positive excursion in filtered Φ_{DP} , followed by a return toward the underlying trend, is characteristic of differential backscatter phase. The segment endpoints therefore provide relatively stable constraints on the net propagation-phase change across the feature.

Figure 8 shows an example of a filtered ray (from S-band), with several oscillating cycles of increasing/decreasing Φ_{DP} . We place a divider at each location where the trend changes from negative to positive. The pink vertical lines show these dividing locations. We then apply the procedure to each segment in turn.

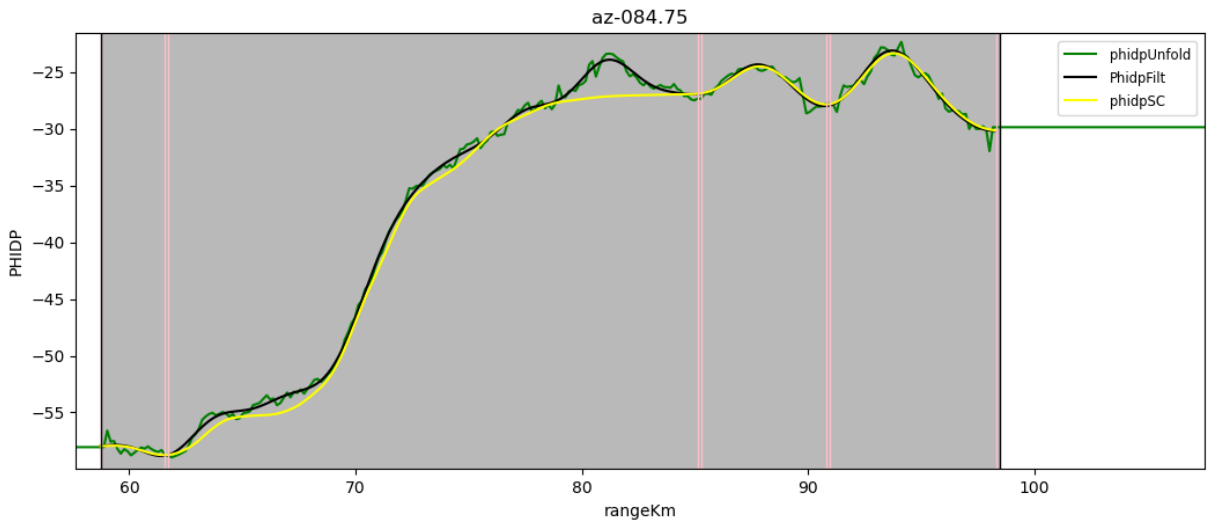


Figure 8: Dividing the Φ_{DP} data into suitable segments.
Black: filtered Φ_{DP} . Yellow: Φ_{DP}^{sc} after applying the self-consistency procedure.

Within each segment, the self-consistency estimate K_{DP}^{sc} is scaled so that its integrated phase change matches the phase difference between the filtered Φ_{DP} values at the two segment endpoints.

Defining

$$\Delta\Phi_{DP} = \Phi_{DP}^f(r_2) - \Phi_{DP}^f(r_1), \quad (12)$$

where Φ_{DP}^f denotes filtered differential phase, the scaled estimate is

$$K_{DP}^{sc}(r) = \frac{\Delta\Phi_{DP}}{2 \int_{r_1}^{r_2} K_{DP}^{sc*}(s) ds} K_{DP}^{sc*}(r), \quad (13)$$

where K_{DP}^{sc*} is obtained from the Z_H - Z_{DR} self-consistency relationship. The self-consistency relation therefore determines the distribution of phase change within the segment, while the measured endpoint values constrain the total phase change.

The backscatter differential phase δ is computed as:

$$\delta = \Phi_{DP} - \Phi_{DP}^{sc}, \delta > 0 \quad (14)$$

We compute the mean δ for the segment. If this mean exceeds a threshold (1.0 degree works well for the cases examined here) we keep the modified Φ_{DP}^{sc} profile. If the mean is below the threshold we leave the Φ_{DP} profile unchanged.

6.4. Applying the self-consistency procedure iteratively

There is some feedback that we need to be aware of. Our attenuation correction is a function of K_{DP} - see Eq. (5) and Eq. (6). But our self-consistency K_{DP}^{sc} uses Z_H and Z_{DR} values that have been corrected for attenuation - see Eq. (11). Therefore we apply the procedure twice in succession, as follows:

- (a) Compute attenuation, and corrected values for Z_h and Z_{dr} , using filtered K_{DP} .
- (b) Compute self-consistency K_{DP}^{sc} .
- (c) Recompute attenuation, using K_{DP}^{sc} instead of K_{DP} .
- (d) Recompute K_{DP}^{sc} .

For the cases tested so far, two passes have been sufficient, with no apparent need for further iteration.

7. Results

7.1. Ray plot results

We will use a C-band case (from the CSWR COW2 radar on 2025/06/18) to illustrate some results.

Figure 9 shows a C-band ray plot for an azimuth 210.5 degrees. Consider the range segment from about 8 km to 33 km. In the right panel the Φ_{DP}^{sc} line (yellow) lies close to the Φ_{DP} line (black). At the top end the yellow line diverges, pointing to the presence of backscatter differential phase. The general agreement between the two lines indicates that the location of the propagation phase shift has not been modified significantly through the application of the self-consistency procedure. This can be explained by reasonably constant reflectivity (red) and Z_{DR} (blue) values across the segment - see left panel.

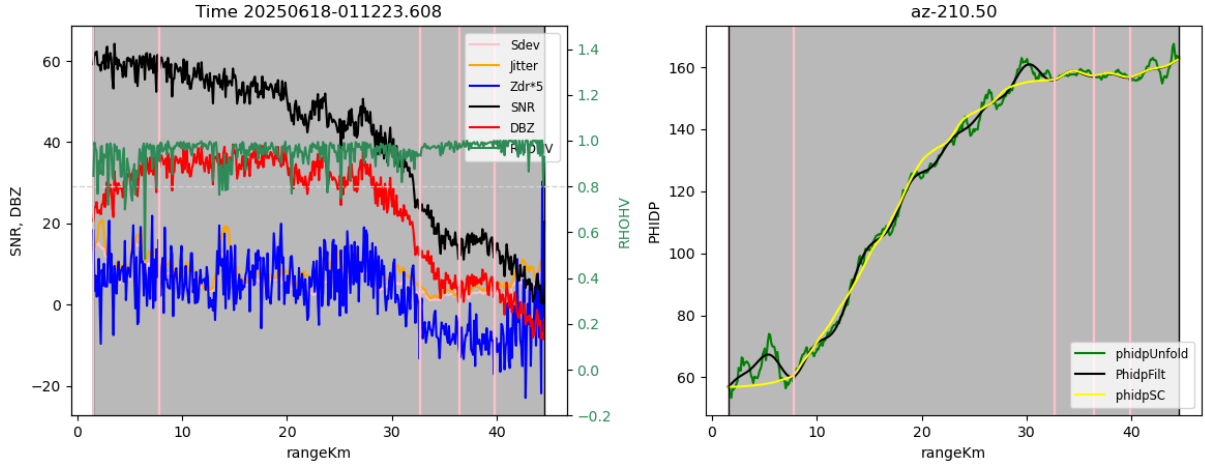


Figure 9: Ray plot, C-band, 2025/06/18 01:12:23 UTC.

Left: radar moments.

Right: Black: filtered Φ_{DP} . Yellow: Φ_{DP}^{sc} after applying the self-consistency procedure.

Figure 10 shows a similar result (for an azimuth 205.5 degrees) but with more backscatter differential phase. The three local maxima in the Φ_{DP} (black) line show the location of the backscatter differential phase, which is corrected by the yellow line for Φ_{DP}^{sc} . The overall shape of the Φ_{DP} line is preserved - see the relatively constant reflectivity and Z_{DR} in the left panel.

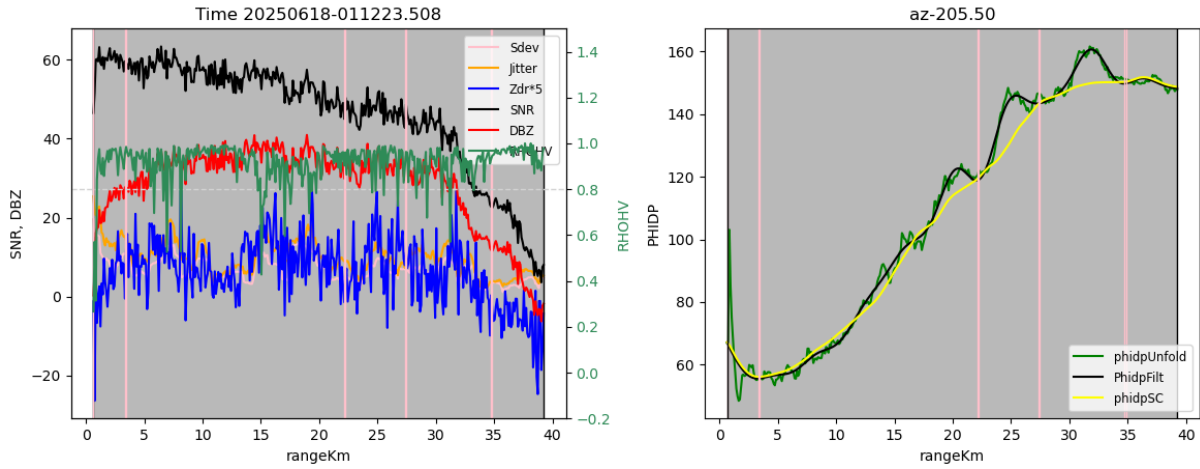


Figure 10: C-band ray example at azimuth 205.5 degrees.

Black: filtered Φ_{DP} . Yellow: Φ_{DP}^{sc} after applying the self-consistency procedure.

Figure 11 is considerably different. In the left half of the plot the yellow Φ_{DP}^{sc} is considerably below the black Φ_{DP} line, while the opposite is true in the right half of the plot. This represents a considerable redistribution of the phase shift locations in the ray.

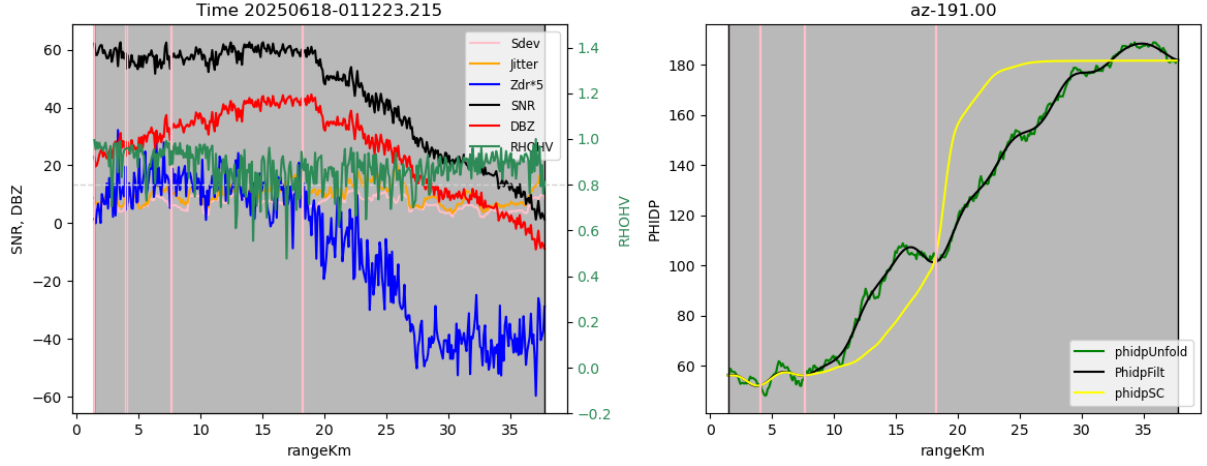
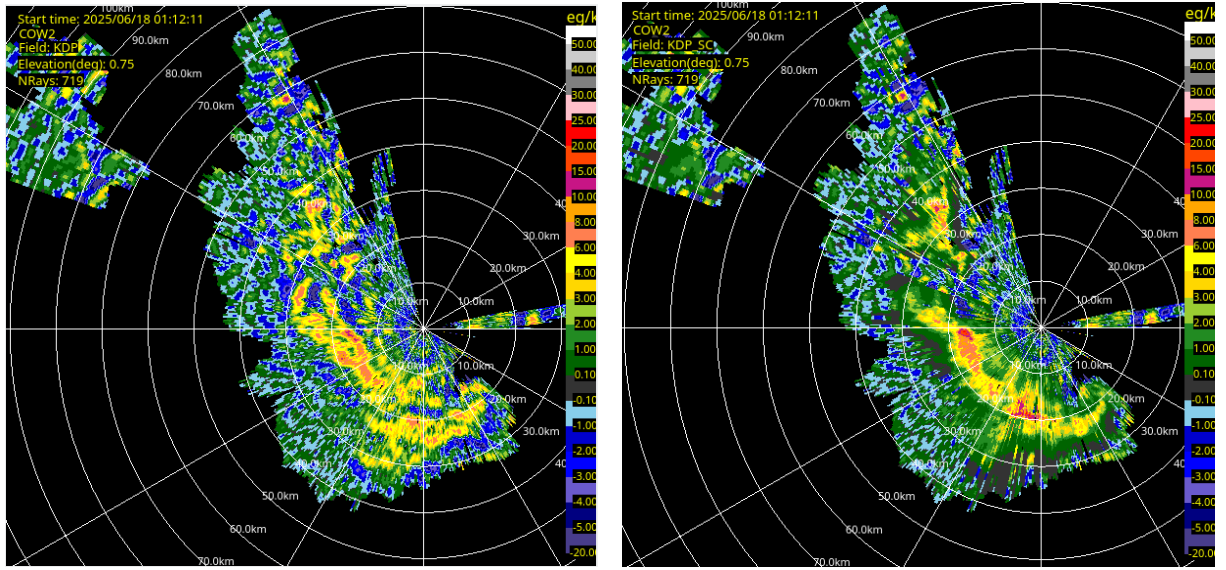


Figure 11: C-band ray example showing substantial redistribution of propagation phase.
Black: filtered Φ_{DP} . Yellow: Φ_{DP}^{sc} after applying the self-consistency procedure.

7.2. PPI plot results

7.2.1 C-band PPIs



(a) PPI plot of K_{DP} , C-band, 2025/06/18

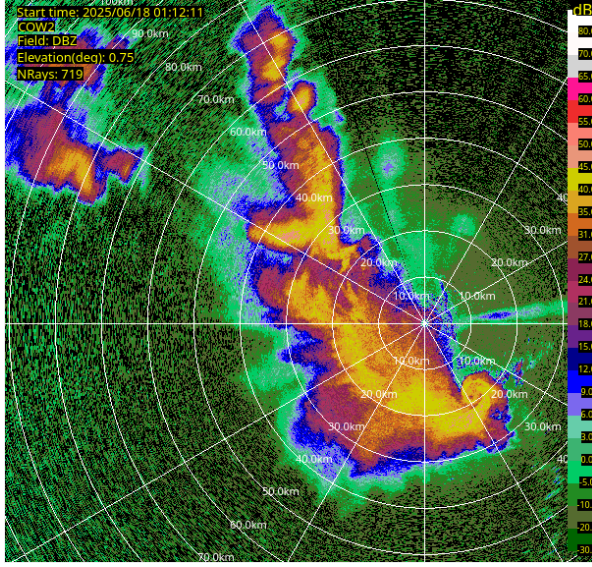
(b) PPI plot of K_{DP}^{sc} , C-band, 2025/06/18.

Figure 12: K_{DP} and K_{DP}^{sc} for C-band case in Figure 1

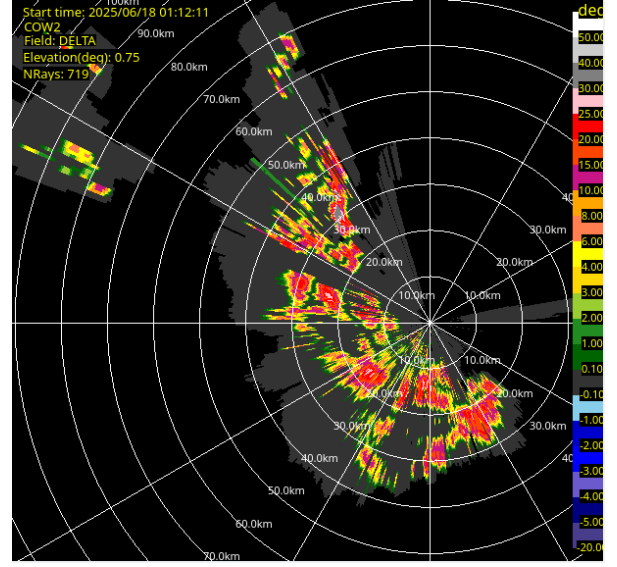
Figure 12 shows the K_{DP} (left panel) and K_{DP}^{sc} (right panel) for the C-band case shown in Figure 1. This illustrates how both the location and strength of the K_{DP} field can change after the application of the self-consistency method.

You can see that the self-consistency procedure that produces K_{DP}^{sc} has changed the shape of the K_{DP} areas, increased the intensity, and moved some of them away from the radar.

Figure 13 shows the backscatter differential phase (δ) field (right panel) for this case, alongside the reflectivity (left panel) for reference.



(a) PPI plot of reflectivity (dBZ), C-band, 2025/06/18



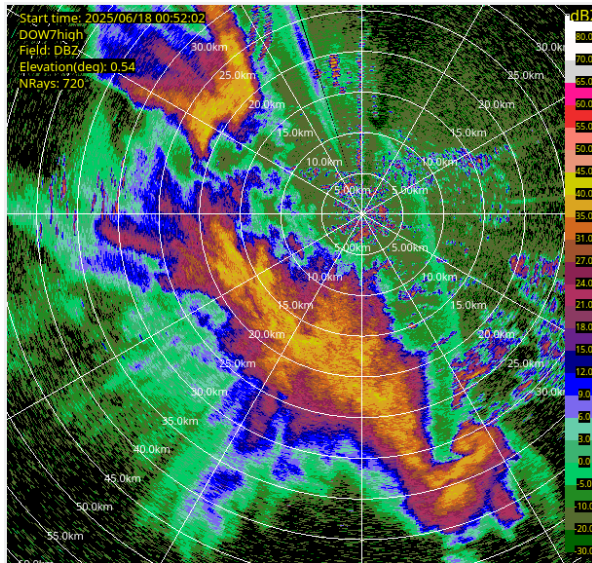
(b) PPI plot of δ , C-band, 2025/06/18.

Figure 13: Reflectivity and δ for C-band case in Figure 12

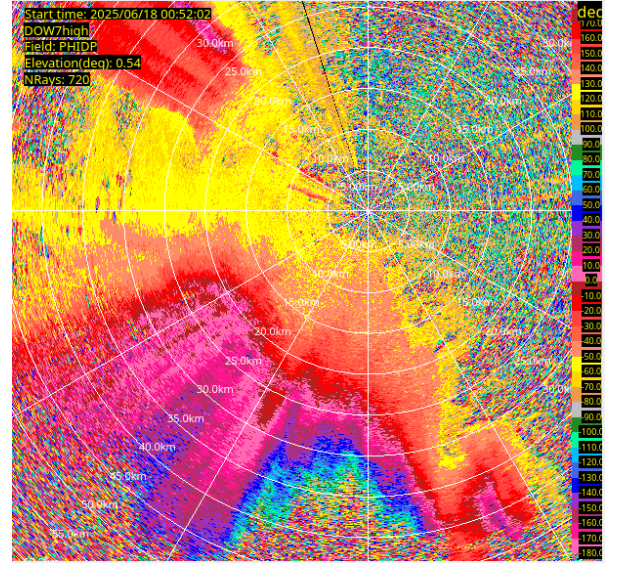
7.2.2 X-band PPIs

The figures below show the same storm as seen from the X-band DOW7 radar.

Figure 14(a) shows the reflectivity, and (b) shows unfiltered Φ_{DP} .

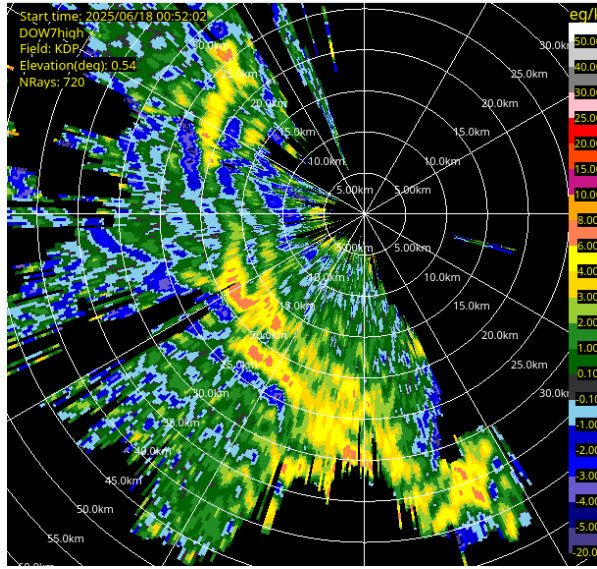


(a) PPI plot of reflectivity (dBZ), X-band, 2025/06/18

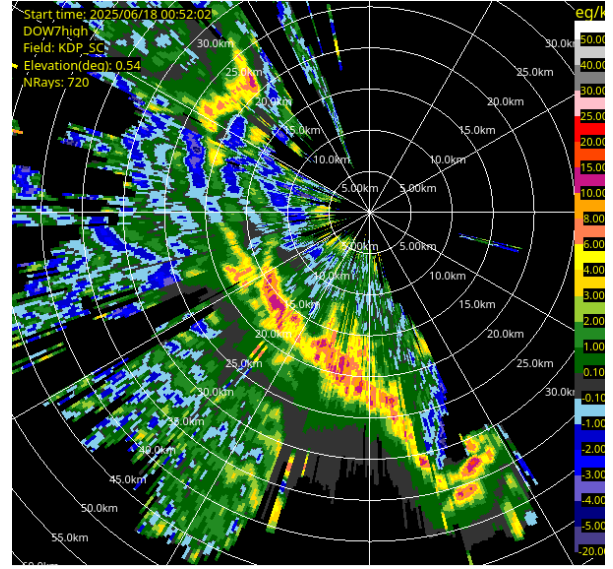


(b) PPI plot of Φ_{DP} , X-band, 2025/06/18.

Figure 14: Reflectivity and Φ_{DP} for X-band case



(a) PPI plot of K_{DP} , X-band, 2025/06/18



(b) PPI plot of K_{DP}^{sc} , X-band, 2025/06/18.

Figure 15: K_{DP} and K_{DP}^{sc} for X-band case

Figure 15(a) shows the K_{DP} and (b) shows K_{DP}^{sc} . This X-band case nicely illustrates how the self-consistency method concentrates the phase shift to the higher-reflectivity areas.

8. Conclusions

We have developed a method that assesses and filters Φ_{DP} , and then applies an endpoint-constrained self-consistency procedure to estimate K_{DP}^{sc} and the associated backscatter differential phase δ .

A key feature of the present method is that the self-consistency procedure is applied to carefully selected segments of the Φ_{DP} range profile. After filtering, we divide the profile into segments that contain a monotonically increasing section followed by a monotonically decreasing section. Within each segment, the self-consistency relationship determines the distribution of propagation phase, while the filtered Φ_{DP} values at the segment endpoints constrain the total phase change. This allows the self-consistency procedure to be applied locally and prevents errors in the self-consistency reconstruction within one segment from propagating into the normalization of adjacent segments. We believe that this local, endpoint-constrained approach may provide a significant advantage in terms of robustness.

We have tested the method on cases from all three major weather radar bands—S, C, and X. Initial testing indicates consistent behavior across the S-, C-, and X-band cases examined here.

The method will be released as part of the open-source LROSE software on GitHub, providing a documented reference implementation for subsequent evaluation and use.

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